

REFERME INVESTMENTS

DATA MODEL

Power BI Semantic Model Design
February - June 2026

Document Control	Details
Document ID	RMI-DM-001
Version	1.0
Purpose	Define the logical Power BI semantic model required to support investment evaluation, KPI analysis, scoring, monitoring and risk reporting.
Model Pattern	Star-schema-oriented semantic model with conformed Business, Date and KPI dimensions plus snapshot / monthly fact tables.
Source Reality	The current project source supports a snapshot evaluation model. Historical monthly raw data required for full trend/monitoring analysis is not established.
Control	Future-ready tables and fields are design recommendations. They must not be represented as existing production data until actual sources are confirmed.

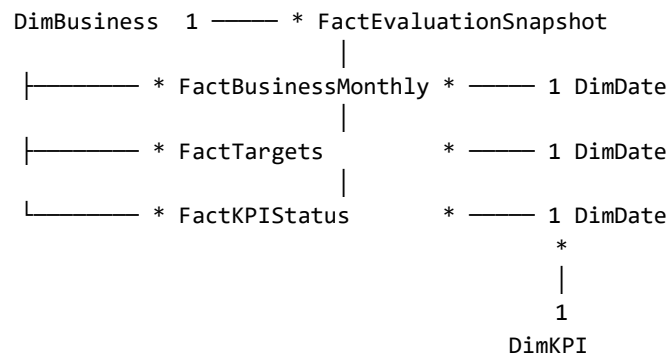
1. Model Objective

The data model provides a consistent analytical layer between source data and the five Power BI pages. Its purpose is to support target-level filtering, KPI calculations, five-dimension scoring, overall Investment Score, recommendation support, portfolio monitoring, risk alerts and visible data-quality status.

2. Current-State vs Future-Ready Design

Layer	Status	What It Supports
Evaluation Snapshot	Supported conceptually now	Business-level snapshot values, five dimension scores, Investment Score and recommendation.
KPI Framework	Supported now	28 KPI names grouped into Financial Strength, Operational Performance, Growth, Customer Health and Risk.
Scoring / Assumptions	Supported as proposed methodology	Dimension weights and configurable scoring/recommendation assumptions; final values require approval.
Business Dimension	Derivable	Clean investment-target master for slicers and relationships.
KPI Dimension	Derivable	KPI catalogue for dimension/KPI navigation and metadata.
Date Dimension	Future-ready / required for history	Required for period trends, CAGR, score changes and portfolio monitoring.
Monthly Business Fact	Future-ready	Required to calculate the KPI framework from period-level raw financial, operational and customer inputs.
Targets Fact	Future-ready	Required for Actual vs Target reporting.
KPI Status Fact	Optional future-ready	Stores KPI value, normalized score, status and approved threshold output at business/date/KPI grain.

3. Recommended Semantic Model



Assumptions = disconnected methodology / parameter table

Recommended relationship behavior is single-direction filtering from dimensions to fact tables. Bi-directional relationships should be avoided unless a specific analytical requirement justifies them.

4. Table Definitions

Table	Grain / Key	Core Columns	Purpose	Status
DimBusiness	One row per BusinessID	BusinessID; Business Type; optional evaluation status / approved descriptive attributes	Conformed business master used for slicers, drill-through and business-level filtering.	Derive from validated business master
DimDate	One row per calendar date	Date; Month; Month Number; Quarter; Year; MonthEndDate	Conformed date dimension for trends, period comparisons and monitoring.	Future-ready / required for time analysis
DimKPI	One row per KPI	KPIKey; KPI Name; Dimension; Unit; Direction; Display Order	Metadata dimension for the 28 KPI catalogue.	Derive from KPI Catalogue
FactEvaluationSnapshot	One row per BusinessID per evaluation snapshot	Revenue; EBITDA; Growth %; Retention %; Risk; five dimension scores; Investment Score; Recommendation	Supports Overview and snapshot Investment Evaluation.	Current conceptual / snapshot layer
FactBusinessMonthly	One row per BusinessID + MonthEndDate	Raw financial, operations and customer inputs required for KPI calculation	Supports full KPI calculation, trends, CAGR, volatility and Business Profile history.	Future-ready
FactTargets	One row per BusinessID + MonthEndDate	Revenue target; EBITDA target; customer / operational targets as approved	Supports actual-vs-target monitoring.	Future-ready
FactKPIStatus	One row per BusinessID + Date + KPIKey	KPI Value; KPI Score; Status; approved threshold outputs	Supports KPI-level drill-down, risk status and explainability.	Optional future-ready
Assumptions	One row per methodology assumption	Assumption Name; Value	Stores configurable scoring weights / thresholds without creating a model relationship.	Disconnected; proposed methodology

5. Relationship Design

From	To	Cardinality	Filter Direction
DimBusiness[BusinessID]	FactEvaluationSnapshot[BusinessID]	1:*	Single
DimBusiness[BusinessID]	FactBusinessMonthly[BusinessID]	1:*	Single
DimDate[Date]	FactBusinessMonthly[MonthEndDate]	1:*	Single
DimBusiness[BusinessID]	FactTargets[BusinessID]	1:*	Single
DimDate[Date]	FactTargets[MonthEndDate]	1:*	Single
DimBusiness[BusinessID]	FactKPIStatus[BusinessID]	1:*	Single
DimDate[Date]	FactKPIStatus[Date]	1:*	Single
DimKPI[KPIKey]	FactKPIStatus[KPIKey]	1:*	Single

6. FactBusinessMonthly - Recommended Input Structure

The source project identifies the logical inputs below as the data needed for full KPI calculation. Exact physical source fields, database columns and data types remain TBD.

Area	Recommended Logical Fields	KPI Use
Financial	Revenue; gross-profit / direct-cost basis; EBITDA; Operating Cash Flow; Operating Cost; Labor Cost	Revenue growth, margins, cash-flow measures, cost ratios and risk calculations.
Operations	Employee Count; Capacity Used; Capacity Available; On-Time Services; Total Services	Revenue per Employee, Capacity Utilization, On-Time Service and operational efficiency.
Customers	Active Customers; New Customers; Lost Customers; Repeat Customers; Complaints; Recurring Revenue; Top-Customer Revenue	Customer growth, acquisition, recurring revenue, ARPC, retention/churn, repeat rate, complaints and concentration.
Identity & Time	BusinessID; MonthEndDate	Business relationships, time trends, CAGR, volatility and period comparison.

7. Model-to-KPI Mapping

KPI Dimension	Primary Model Sources
Financial Strength	FactBusinessMonthly financial inputs; DimDate for growth/time comparison; scoring metadata where approved.
Operational Performance	FactBusinessMonthly workforce/operations inputs.
Growth	FactBusinessMonthly revenue/customer/recurring-revenue inputs plus DimDate for CAGR and period change.
Customer Health	FactBusinessMonthly customer inputs and customer-level concentration evidence where available.
Risk	FactBusinessMonthly + time history + concentration data + data-quality outputs; FactKPIStatus where implemented.

8. Scoring Model Integration

- KPI values should be calculated from validated source inputs before normalization.
- Normalized KPI scores should use approved scoring direction, thresholds and benchmarks.
- Dimension scores should be calculated from the approved KPI-to-dimension weighting logic.
- The overall Investment Score should use the validated dimension weights; the current proposed model is 25% / 20% / 20% / 15% / 20%.
- Recommendation logic should remain parameterized until Invest / Review / Reject thresholds are approved.
- The Assumptions table should remain disconnected so methodology values can be controlled without causing unintended filtering.

9. Data Model Support by Power BI Page

Power BI Page	Primary Tables	Current / Future Support
Investment Overview	DimBusiness + FactEvaluationSnapshot + Assumptions	Snapshot support available conceptually; recommendation depends on approved rules.
Investment Evaluation	DimBusiness + FactEvaluationSnapshot; later FactKPIStatus + DimKPI	Snapshot scores supported; full KPI evidence improves when KPI-status detail exists.
Business Profile	DimBusiness + FactBusinessMonthly + DimDate	Full trend/profile design requires monthly history; snapshot fallback is limited.
Portfolio Monitoring	DimBusiness + FactBusinessMonthly + FactTargets + DimDate + scoring history	Requires historical actuals / targets; not supported by a single snapshot alone.
Risk & Alerts	FactBusinessMonthly + FactKPIStatus + DimKPI + DimDate	Full alerting requires approved rules and supporting history; snapshot fallback may be limited.

10. Power Query / Data Preparation Requirements

- Create DimBusiness by referencing validated business data and removing duplicate BusinessID records.
- Create DimKPI from the KPI Catalogue; assign a stable KPIKey and display order.
- Use a proper DimDate table before enabling time-intelligence analysis.
- Keep the evaluation snapshot as a fact table rather than adding business attributes repeatedly across derived tables.
- When monthly raw data becomes available, append at BusinessID + MonthEndDate grain rather than adding one column per month.
- Standardize percentage, score and currency data types in the semantic layer.
- Apply data-quality checks before score/recommendation measures are treated as decision-ready.

11. Modeling Controls

Control	Requirement
Grain	Define and preserve the grain of every fact table before relationships/measures are created.
Keys	BusinessID, Date and KPIKey must be stable and unique on their respective dimensions.
Relationships	Prefer 1:* single-direction dimension-to-fact relationships.
Many-to-many	Avoid unless a validated business requirement requires a bridge design.
History	Do not create trend calculations from a single snapshot.
Targets	Do not display Actual vs Target until approved target data exists.
Scoring	Do not hard-code unapproved methodology values as final business rules.
Data Quality	Keep data-quality limitations visible to downstream measures and reports.

12. Open Technical / Business Decisions

- Confirmed production source systems and ingestion mechanism.
- Actual field names, physical data types, keys and source ownership.
- Monthly / daily evaluation grain for future history.
- Historical data availability and retention period.
- Target-setting process and target data ownership.
- Final KPI direction, normalization benchmarks and threshold rules.
- Final recommendation thresholds and scoring weights.
- Refresh frequency, security / RLS, deployment and SLA.